

Linearized Drift-Pressure Model: Definitions, Equilibrium, and the Linear Stability Matrix

Overview

This note collects the working derivation into a single, self-contained reference. Section 1 fixes notation and lists every constant/operator that appears, with its physical meaning and (where defined) its formula. Section 2 states the two governing equations we start from. Section 3 derives the axisymmetric, steady equilibrium. Section 4 introduces the perturbation ansatz and linearizes both equations term by term. Section 5 assembles the result into the 2×2 linear operator (eigenvalue) problem. Section 6 collects, in one place, every approximation and open question flagged in the original derivation, so the assumptions underlying the final matrix are explicit and auditable.

1 Notation, Operators, and Constants

1.1 Bracket and Laplacian operators

Generalized Poisson bracket. For two scalar fields $a(r, \theta)$, $b(r, \theta)$,

$$\{a, b\} \equiv a_x b_y - a_y b_x, \quad (1)$$

where subscripts denote partial derivatives. In polar coordinates this is written explicitly as

$$\{a, b\} = \frac{1}{r} (a_r b_\theta - a_\theta b_r). \quad (2)$$

Laplacian.

$$\Delta \equiv \nabla^2. \quad (3)$$

When axisymmetry is assumed (no θ -dependence in the background), the full Laplacian acting on a perturbation $\propto e^{im\theta + \gamma t}$ separates as

$$\Delta \rightarrow \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}, \quad (4)$$

and we define the *m-th harmonic radial Laplacian operator*

$$\Delta_m \equiv \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) - \frac{m^2}{r^2}, \quad (5)$$

i.e. $\Delta(\hat{\varphi} e^{im\theta + \gamma t}) = \Delta_m \hat{\varphi} e^{im\theta + \gamma t}$, since $\partial_\theta^2 \rightarrow -m^2$ on this harmonic.

1.2 Field variables

Symbol	Meaning
φ	Normalized plasma potential, $\varphi = \phi/\phi_\star$, where $\phi_\star$ is the (central) amplitude of the applied wall potential.
P	Normalized pressure, $P = p/p_0$, where p_0 is the plasma pressure on axis at the trap midplane.
ρ	Mass density, $\rho = m_i n_0$ (ion mass times equilibrium number density).

1.3 Constants and coefficients

Symbol	Definition	Physical meaning
U	$U = \frac{p_{0i}}{e n \phi}$	Ratio of ion pressure to (number density $\times$ potential); governs finite-Larmor-radius (FLR) effects.
H	$H = \frac{\tau^2 e \phi}{c L T_e} \left(\frac{J_{i0}}{B} \right)_{\text{wall}} \frac{1}{\langle \rho B^{-2} \rangle}$	Plasma–wall coupling strength (line-tying effect).
$\mathcal{K}$	$\mathcal{K} = 2 P_0 \tau^2 \frac{\langle q(z) \alpha(z) B^{-1} \rangle}{\langle \rho B^{-2} \rangle}$	Normalized average magnetic curvature.
J_{i0}	—	Equilibrium axial ion current.
τ	$\tau = \frac{B R^2}{c \phi}$	$E \times B$ drift period (BR^2 is evaluated on each field line).
L	—	Half-length of the trap.
$q(z)$	$\langle q \rangle = 1$	Axial pressure profile.
$\alpha(z)$	$\frac{1}{B(r, z)} = \frac{1}{B(0, z)} + \alpha(z) r^2$	Paraxial field expansion coefficient; obtainable from $\nabla \cdot \mathbf{B} = 0$ together with the known on-axis field $B(0, z)$.
T_e	—	Electron temperature.
ϕ_{fe}	—	Floating potential (flat in radius, evaluated at constant temperature).
ϕ_{wall}	—	End-wall bias (applied externally).
φ_w	$\varphi_w = \phi_{fe} + \phi_{\text{wall}}$	Total effective wall potential.
$\nu_{\perp}^P$	—	Collisional transverse (cross-field) diffusion rate acting on P .
$\nu_{\parallel}^P$	—	Axial diffusion rate acting on P .
$\nu_{\perp}, \nu_{\parallel}$	—	Transverse/axial diffusion rates acting on φ (analogous role to $\nu_{\perp}^P, \nu_{\parallel}^P$ in the pressure equation; not separately defined in the source material).
Q_P	$Q_P \approx Q_P(r)$	Power source term in the pressure equation.

Open notational issue (flagged in original notes): it is not fully specified whether ϕ_w is normalized with respect to ϕ in the same way as φ , i.e. whether $\varphi_w \equiv \phi_w / \phi_{\star}$ consistently. This ambiguity is carried through the equilibrium relation $\varphi_0 \approx \varphi_w$ in Section 3.

2 Governing Equations

The starting point is a pair of coupled evolution equations for the normalized potential φ and normalized pressure P :

$$\partial_t \Delta \varphi + \{\varphi, \Delta \varphi\} - U \nabla \cdot \{\nabla \varphi, P\} = H \cdot (\varphi - \varphi_w) + \mathcal{K} \{P, r^2\} + \nu_\perp \Delta^2 \varphi - \nu_\parallel \Delta \varphi - U \Delta S_P, \quad (6)$$

$$\partial_t P + \{\varphi, P\} = \nu_\perp^P \Delta P - \nu_\parallel^P P + Q_P. \quad (7)$$

Equation (6) is the *vorticity (potential) equation*; Equation (7) is the *pressure equation*. Together they form the closed system that is linearized below.

3 Equilibrium

Assumption E1 (axisymmetric, steady equilibrium). The background state has no θ -dependence and is time-independent: $\partial_t(\cdot)_0 = 0$ and $\partial_\theta(\cdot)_0 = 0$.

Under Assumption E1, any bracket of two equilibrium quantities vanishes, $\{\varphi_0, \Delta \varphi_0\} = 0$, since $\{a, b\}$ in Eq. (2) requires a θ -derivative of at least one field. The $\nabla \cdot \{\nabla \varphi, P\}$ term also drops at equilibrium for the same reason. Equation (6) therefore collapses to

$$0 = H \cdot (\varphi_0 - \varphi_w) + \nu_\perp \Delta^2 \varphi_0 - \nu_\parallel \Delta \varphi_0. \quad (8)$$

Assumption E2. The diffusive terms $\nu_\perp \Delta^2 \varphi_0$ and $\nu_\parallel \Delta \varphi_0$ are small compared to the line-tying term $H(\varphi_0 - \varphi_w)$.

With Assumption E2, the equilibrium relation reduces to

$$H(\varphi_0 - \varphi_w) = 0 \quad \implies \quad \boxed{\varphi_0 \approx \varphi_w}. \quad (9)$$

That is, to leading order the equilibrium plasma potential simply matches the (effective) wall potential. Two open questions are flagged here in the original notes and remain unresolved approximations of the model (see also Section 6):

- Is it valid to assume $\nu_\parallel \approx \nu_\parallel^P$ (i.e. that the axial diffusion rates for φ and P coincide)?
- Is it valid to assume $\nu_\perp \ll \nu_\parallel$?

3.1 Equilibrium rotation frequency

The $E \times B$ drift velocity is

$$\mathbf{v}_E = \frac{c \hat{\mathbf{b}} \times \nabla \varphi}{B}. \quad (10)$$

Since the equilibrium potential has no θ -dependence, $\nabla \varphi_0 = \hat{r} \frac{d\varphi_0}{dr}$, so the equilibrium $E \times B$ flow is purely azimuthal and defines an equilibrium rotation frequency $\Omega_0(r)$ through

$$\Omega_0(r) \equiv \frac{1}{r} \frac{\partial \varphi_0}{\partial r} \propto \frac{1}{r} \frac{\partial \varphi_w}{\partial r}, \quad (11)$$

using $\varphi_0 \approx \varphi_w$ from Eq. (9) in the last step.

4 Perturbation Ansatz and Linearization

Assumption L1 (normal-mode ansatz). Both fields are perturbed by a single azimuthal/temporal Fourier harmonic, with small complex amplitudes $\widehat{\varphi}(r)$, $\widehat{\rho}(r)$:

$$\varphi(r, \theta, t) = \varphi_0(r) + \widehat{\varphi}(r) e^{im\theta + \gamma t}, \quad (12)$$

$$P(r, \theta, t) = P_0(r) + \widehat{\rho}(r) e^{im\theta + \gamma t}. \quad (13)$$

Here m is the (integer) azimuthal mode number and γ is the (complex) growth rate/frequency.

Assumption L2 (linearization). Only first-order terms in the perturbation amplitudes ($\widehat{\varphi}$, $\widehat{\rho}$) are retained; products of two perturbed quantities are dropped.

The remainder of this section substitutes the ansatz (12)–(13) into Eqs. (6)–(7) and linearizes each term.

4.1 Linearized pressure equation

Starting from Eq. (7),

$$\partial_t P + \{\varphi, P\} = \nu_{\perp}^P \Delta P - \nu_{\parallel}^P P + Q_P, \quad (14)$$

substitute the ansatz and keep only first-order (linear) terms (Assumption L2). The bracket splits into an equilibrium piece (zero, by Assumption E1) and two cross terms:

$$\{\varphi_0, \delta P\} = \frac{1}{r} \frac{\partial \varphi_0}{\partial r} \frac{\partial}{\partial \theta} (\delta P) = im \Omega_0(r) \widehat{\rho} e^{im\theta + \gamma t}, \quad (15)$$

$$\{\delta \varphi, P_0\} = 0 - \frac{1}{r} \frac{\partial}{\partial \theta} (\delta \varphi) \frac{\partial P_0}{\partial r} = -\frac{im}{r} \widehat{\varphi} e^{im\theta + \gamma t} P_0'(r), \quad (16)$$

where $\delta \varphi \equiv \widehat{\varphi} e^{im\theta + \gamma t}$, $\delta P \equiv \widehat{\rho} e^{im\theta + \gamma t}$, and a prime denotes d/dr . The full linearized pressure balance reads

$$\frac{\partial}{\partial t} (P_0 + \delta P) + im \Omega_0 \widehat{\rho} e^{im\theta + \gamma t} - \frac{im}{r} P_0'(r) \widehat{\varphi} e^{im\theta + \gamma t} = -\nu_{\parallel}^P \left(P_0 + \widehat{\rho} e^{im\theta + \gamma t} \right) + \underbrace{\nu_{\perp}^P \Delta P}_{\rightarrow 0}. \quad (17)$$

Assumption L3. The collisional transverse-diffusion contribution $\nu_{\perp}^P \Delta P$ is dropped at this order (set to zero) in the perturbed pressure balance.

Order-zero (equilibrium) balance. Collecting terms with no $e^{im\theta + \gamma t}$ dependence gives the equilibrium condition

$$\underbrace{\{\varphi_0, P_0\}}_{=0 \text{ (Assumption E1)}} = \nu_{\parallel}^P P_0 \implies \nu_{\parallel}^P P_0 = 0 \text{ at equilibrium.} \quad (18)$$

Order-one (perturbed) balance. Collecting the terms proportional to $e^{im\theta + \gamma t}$ and writing $\partial_t \rightarrow \gamma$:

$$\gamma \widehat{\rho} + im \Omega_0 \widehat{\rho} - \frac{im}{r} P_0'(r) \widehat{\varphi} = -\nu_{\parallel}^P \widehat{\rho}. \quad (19)$$

Solving for $r\hat{\rho}$ (multiplying through by r) gives the first row of the eigenvalue system:

$$\boxed{\gamma r \hat{\rho} = im P'_0(r) \hat{\varphi} - r \nu_{\parallel}^P \hat{\rho} - im r \Omega_0 \hat{\rho}} \quad (20)$$

which is more cleanly written as

$$\gamma r \hat{\rho} = im P'_0(r) \hat{\varphi} - \left(\nu_{\parallel}^P + im \Omega_0 \right) r \hat{\rho}. \quad (21)$$

This identifies the two operator entries coupling row 2 of the matrix (acting on the pair $(\Delta_m \hat{\varphi}, \hat{\rho})$, see Section 5):

$$\mathcal{L}_{21} = \frac{im}{r} P'_0(r) \Delta_m^{-1}, \quad (22)$$

$$\mathcal{L}_{22} = -\nu_{\parallel}^P - im \Omega_0. \quad (23)$$

The factor of Δ_m^{-1} in $\mathcal{L}_{21}$ appears because the matrix problem (Section 5) is posed on the variable pair $(\Delta_m \hat{\varphi}, \hat{\rho})$ rather than $(\hat{\varphi}, \hat{\rho})$ directly; $\mathcal{L}_{21}$ converts $\Delta_m \hat{\varphi}$ back to $\hat{\varphi}$ before multiplying by $P'_0(r)$.

4.2 Linearized vorticity equation

Now linearize Eq. (6),

$$\partial_t \Delta \varphi + \{\varphi, \Delta \varphi\} - U \nabla \cdot \{\nabla \varphi, P\} = H(\varphi - \varphi_w) + \mathcal{K} \{P, r^2\} + \nu_{\perp} \Delta^2 \varphi - \nu_{\parallel} \Delta \varphi - U \Delta S_P. \quad (24)$$

Assumption L4. The hyper-diffusive term $\nu_{\perp} \Delta^2 \varphi$ and the source-gradient term $U \Delta S_P$ are dropped (ignored) in the linear stability analysis.

Assumption L5. The background (θ -independent) perturbed-equilibrium quantities are also assumed to have no θ -dependence in φ and P beyond what is captured by the ansatz; i.e. we work strictly to first order in $\hat{\varphi}, \hat{\rho}$ and discard any second-order-in-perturbation (“frightening”) term that would arise from a product of two perturbations.

Laplacian on a single harmonic. Acting on either $\delta \varphi$ or $\delta P \propto e^{im\theta + \gamma t}$, the angular part of Δ produces $-m^2/r^2$, motivating the definition of Δ_m in Eq. (5):

$$\gamma \Delta_m \left(\hat{\varphi} e^{im\theta + \gamma t} \right) \text{ is the linearized form of } \partial_t \Delta \varphi. \quad (25)$$

Term 1: $\{\varphi, \Delta \varphi\}$. Expand $\varphi = \varphi_0 + \delta \varphi$ so that $\Delta \varphi = \Delta \varphi_0 + \Delta_m \delta \varphi$. The bracket splits into three pieces, of which the equilibrium–equilibrium piece vanishes by Assumption E1:

$$\{\varphi_0, \Delta \varphi_0\} = 0, \quad (26)$$

$$\{\delta \varphi, \Delta \varphi_0\} = -\frac{im}{r} \hat{\varphi} e^{im\theta + \gamma t} (\Delta \varphi_0)', \quad (27)$$

$$\{\varphi_0, \Delta \delta \varphi\} = \frac{1}{r} \varphi'_0(r) \Delta_m \left(\hat{\varphi} e^{im\theta + \gamma t} \right) im \equiv im \Omega_0 \Delta_m \left(\hat{\varphi} e^{im\theta + \gamma t} \right), \quad (28)$$

using the definition of $\Omega_0(r)$ from Eq. (11) in the last line.

Term 2: $U \nabla \cdot \{\nabla \varphi, P\}$. Using $\nabla A = \hat{r} \partial_r A + \hat{\theta} \frac{1}{r} \partial_\theta A + \hat{z} \partial_z A$ and $\nabla \cdot \mathbf{A} = \frac{1}{r} \partial_r (r A_r) + \frac{1}{r} \partial_\theta A_\theta + \partial_z A_z$, the two contributing cross terms are

$$\nabla \cdot \{\nabla \varphi_0, \delta P\} = \frac{1}{r} \frac{\partial}{\partial r} \left[r \left(\frac{1}{r} \frac{\partial \varphi_0}{\partial r} im \hat{\rho} e^{im\theta + \gamma t} \right) \right], \quad (29)$$

$$\nabla \cdot \{\nabla \delta \varphi, P_0\} = \frac{1}{r} \frac{\partial}{\partial r} \left[r \left(-\frac{im}{r} P'_0(r) \hat{\varphi} e^{im\theta + \gamma t} \right) \right] + \frac{1}{r} \frac{\partial}{\partial \theta} \left[\frac{1}{r^2} m^2 P'_0(r) \hat{\varphi} e^{im\theta + \gamma t} \right], \quad (30)$$

which together combine to

$$\nabla \cdot \{\nabla \varphi, P\} \Big|_{\text{linear}} = \left[\frac{1}{r} \frac{\partial}{\partial r} (im \hat{\rho} \varphi''_0 - im P'_0 \hat{\varphi}') + \frac{im^3}{r^3} P'_0 \hat{\varphi} \right] e^{im\theta + \gamma t}. \quad (31)$$

Term 3: $\mathcal{K} \{P, r^2\}$. Since r^2 has no θ -dependence, only $\{\delta P, r^2\}$ survives:

$$\mathcal{K} \{P, r^2\} \Big|_{\text{linear}} = -\frac{\mathcal{K}}{r} im \hat{\rho} e^{im\theta + \gamma t} (2r) = -2im \mathcal{K} \hat{\rho} e^{im\theta + \gamma t}. \quad (32)$$

Assembling the linearized vorticity equation. Collecting Terms 1–3 together with the line-tying term $H(\varphi - \varphi_w) \rightarrow H \hat{\varphi} e^{im\theta + \gamma t}$ (using $\varphi_0 \approx \varphi_w$, Eq. (9)) and the axial-diffusion term $-\nu_{\parallel} \Delta_m \hat{\varphi} e^{im\theta + \gamma t}$:

$$\begin{aligned} & \gamma \Delta_m \hat{\varphi} e^{im\theta + \gamma t} + im \Omega_0 \Delta_m \hat{\varphi} e^{im\theta + \gamma t} - \frac{im}{r} \hat{\varphi} e^{im\theta + \gamma t} (\Delta \varphi_0)' \\ & - U \left[\frac{1}{r} \frac{\partial}{\partial r} (im \hat{\rho} \varphi''_0 - im P'_0 \hat{\varphi}') + \frac{im^3}{r^3} P'_0 \hat{\rho} \right] e^{im\theta + \gamma t} \\ & = H \hat{\varphi} e^{im\theta + \gamma t} - 2im \mathcal{K} \hat{\rho} e^{im\theta + \gamma t} - \nu_{\parallel} \Delta_m \hat{\varphi} e^{im\theta + \gamma t}. \end{aligned} \quad (33)$$

Assumption L6. In the term $\frac{im}{r} \hat{\varphi} (\Delta \varphi_0)'$, we set $\Delta \varphi_0 = 0$ (i.e. neglect the equilibrium vorticity itself in this particular cross term). This is acknowledged as somewhat inconsistent with keeping $\Omega_0(r)$ finite elsewhere, but is necessary to close the eigenvalue problem in the form below, given that $\varphi_0 \sim \varphi_w$ already enters through the boundary/line-tying term.

Dividing through by $e^{im\theta + \gamma t}$ and applying Assumption L6, Eq. (33) becomes

$$\begin{aligned} \gamma \Delta_m \hat{\varphi} = & -im \Omega_0 \Delta_m \hat{\varphi} + \frac{im}{r} (U \varphi'_0)' \hat{\varphi} - \frac{imU}{r} \frac{\partial}{\partial r} (P'_0 \hat{\varphi}') + \frac{im^3}{r^3} U P'_0 \hat{\varphi} \\ & + H \hat{\varphi} - \nu_{\parallel} \Delta_m \hat{\varphi} + \frac{imU}{r} \frac{\partial}{\partial r} (\varphi''_0 \hat{\rho}) - 2im \mathcal{K} \hat{\rho}. \end{aligned} \quad (34)$$

5 The Linear System

Equation (34) is linear and homogeneous in $(\Delta_m \hat{\varphi}, \hat{\rho})$, so it defines the first row of a 2×2 operator matrix; Eq. (21) (Section 4.1) defines the second row, after applying Δ_m^{-1} to convert the $\hat{\varphi}$ coefficient

into one acting on $\Delta_m \hat{\varphi}$. The resulting entries are:

$$\mathcal{L}_{11} = -im \Omega_0 + \frac{im}{r} (\Delta\varphi_0)' \Delta_m^{-1} - \frac{imU}{r} \frac{\partial}{\partial r} (P_0' \Delta_m^{-1}) + \frac{im^3}{r^3} U P_0' \Delta_m^{-1} + H \Delta_m^{-1} - \nu_{\parallel}, \quad (35)$$

$$\mathcal{L}_{12} = \frac{imU}{r} \frac{\partial}{\partial r} (\varphi_0'') - 2im \mathcal{K}, \quad (36)$$

$$\mathcal{L}_{21} = \frac{im}{r} P_0'(r) \Delta_m^{-1}, \quad (37)$$

$$\mathcal{L}_{22} = -\nu_{\parallel}^P - im \Omega_0. \quad (38)$$

(In Eq. (35), the term $\frac{im}{r} (\Delta\varphi_0)' \Delta_m^{-1}$ is formally present from Eq. (33) before Assumption L6 sets $\Delta\varphi_0 = 0$; it is retained here for completeness/traceability but evaluates to zero under that assumption.)

The full linear eigenvalue problem is then

$$\begin{pmatrix} \mathcal{L}_{11} & \mathcal{L}_{12} \\ \mathcal{L}_{21} & \mathcal{L}_{22} \end{pmatrix} \begin{pmatrix} \Delta_m \hat{\varphi} \\ \hat{\rho} \end{pmatrix} = \gamma \begin{pmatrix} \Delta_m \hat{\varphi} \\ \hat{\rho} \end{pmatrix}. \quad (39)$$

This is a generalized eigenvalue problem for the complex growth rate γ , with eigenvector $(\Delta_m \hat{\varphi}, \hat{\rho})^T$. Unstable modes correspond to $\text{Re}(\gamma) > 0$.

6 Summary of Approximations and Open Questions

This section gathers, in one place, every approximation, assumption, and unresolved question flagged during the derivation. They are grouped by where they enter.

6.1 Equilibrium approximations

- **(E1) Axisymmetric, steady background:** $\partial_{\theta}(\cdot)_0 = 0$ and $\partial_t(\cdot)_0 = 0$. This kills every equilibrium-equilibrium Poisson bracket, e.g. $\{\varphi_0, \Delta\varphi_0\} = 0$.
- **(E2) Diffusion subdominant at equilibrium:** $\nu_{\perp} \Delta^2 \varphi_0$ and $\nu_{\parallel} \Delta \varphi_0$ are dropped relative to $H(\varphi_0 - \varphi_w)$, giving the simple closure $\varphi_0 \approx \varphi_w$.
- *Open question:* is it valid to assume $\nu_{\parallel} \approx \nu_{\parallel}^P$ (equal axial diffusion rates for potential and pressure)?
- *Open question:* is it valid to assume $\nu_{\perp} \ll \nu_{\parallel}$ (transverse diffusion negligible relative to axial)?
- *Open notational question:* is φ_w normalized consistently with φ (i.e. both divided by the same reference $\phi_{\star}$)? The source material flags this as unresolved.
- *Open simplification under consideration:* can the axial pressure profile be approximated as uniform, $q(z) = 1$, given that $\langle q \rangle = 1$ by definition?

6.2 Perturbation ansatz and linearization

- **(L1) Single-harmonic normal-mode ansatz:** $\varphi = \varphi_0(r) + \hat{\varphi}(r) e^{im\theta + \gamma t}$, and similarly for P , with m an integer azimuthal mode number and γ the complex growth rate.
- **(L2) Strict linearization:** only terms first-order in $(\hat{\varphi}, \hat{\rho})$ are kept; all quadratic-or-higher perturbation products are dropped.

- **(L3) Transverse diffusion dropped in the pressure equation:** $\nu_{\perp}^P \Delta P \rightarrow 0$ in the linearized pressure balance.
- **(L4) Two terms dropped outright in the vorticity equation:** the hyper-diffusive term $\nu_{\perp} \Delta^2 \varphi$, and the source term $U \Delta S_P$ — both ignored for the linear stability analysis.
- **(L5) No second-order ("frightening") terms:** any term that would multiply two perturbed quantities together is discarded as part of strict linearization.
- **(L6) $\Delta \varphi_0 = 0$ in one specific cross term:** when simplifying $\frac{im}{r} \widehat{\varphi} (\Delta \varphi_0)'$ in the assembled vorticity equation, the equilibrium vorticity $\Delta \varphi_0$ is set to zero. This is flagged explicitly in the original derivation as "*a little inconsistent*" with treating $\Omega_0(r)$ as generally nonzero, but is retained as necessary to close the eigenvalue problem, leaning on $\varphi_0 \approx \varphi_w$ already supplying the relevant equilibrium information through the H term.

6.3 Operator-level approximations

- **Angular part of the Laplacian on a single harmonic:** $\partial_{\theta}^2 \rightarrow -m^2$ whenever acting on $\delta \varphi$ or δP , which is what motivates defining $\Delta_m \equiv \frac{1}{r} \partial_r (r \partial_r) - m^2 / r^2$ in place of the full Δ .
- **Matrix posed on $(\Delta_m \widehat{\varphi}, \widehat{\rho})$, not $(\widehat{\varphi}, \widehat{\rho})$:** consequently $\mathcal{L}_{11}$ and $\mathcal{L}_{21}$ each carry a trailing Δ_m^{-1} to convert back from $\Delta_m \widehat{\varphi}$ to $\widehat{\varphi}$ where needed.

6.4 Quantities defined only implicitly or left undefined in the source material

- $\nu_{\perp}$ and $\nu_{\parallel}$ (the diffusion rates acting on φ in the vorticity equation) are used throughout but never explicitly defined, in contrast to their pressure-equation counterparts $\nu_{\perp}^P, \nu_{\parallel}^P$, which are identified as collisional transverse diffusion and axial diffusion, respectively.
- $\alpha(z)$, the paraxial field expansion coefficient, is defined implicitly via $\frac{1}{B(r,z)} = \frac{1}{B(0,z)} + \alpha(z)r^2$ and must be solved for using $\nabla \cdot \mathbf{B} = 0$ together with the known on-axis profile $B(0, z)$, rather than from a closed-form expression.